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MTH622 Vectors and Classical Mechanics

Question No : 39 of 41

Find the moments of inertia of a ring of radius $2a$ about an axis through center.

Answer (Please click here to Add Answer)

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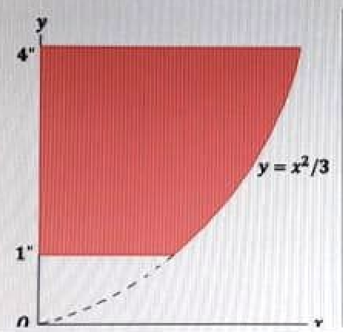
MTH622 Vectors and Classical Mechanics MC18

Question No : 41 of 41 **Marks: 5 (Budgeted Time 10 Min)**

Calculate the moment of inertia of the shaded area given in figure about y-axis.



Answer (Please click here to Add Answer)

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MTH622 Vectors and Classical Mechanics

Question No : 40 of 41

Marks: 5 (Budgeted Time 10 Min)

Calculate the moment of inertia of a uniform rod of length L about an axis perpendicular to the rod and passing through an end point.

(Let the x-axis be chosen along the length of the rod, with origin at one end point Let *M* and *a* be the mass and length of rod respectively and suppose the rod to be composed of small elements.)

Answer (Please click here to Add Answer)

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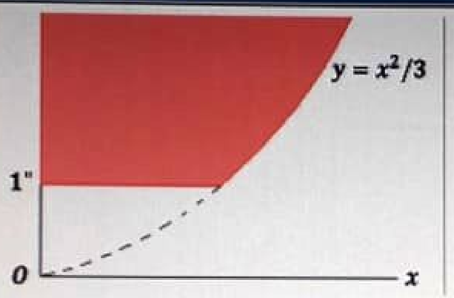
MTH622 Vectors and Classical Mechanics

Question No : 41 of 41 **Marks: 5 (Bu**



whereas an elementary strip from the shaded area whose Moment of inertia is

Answer (Please click here to Add Answer)

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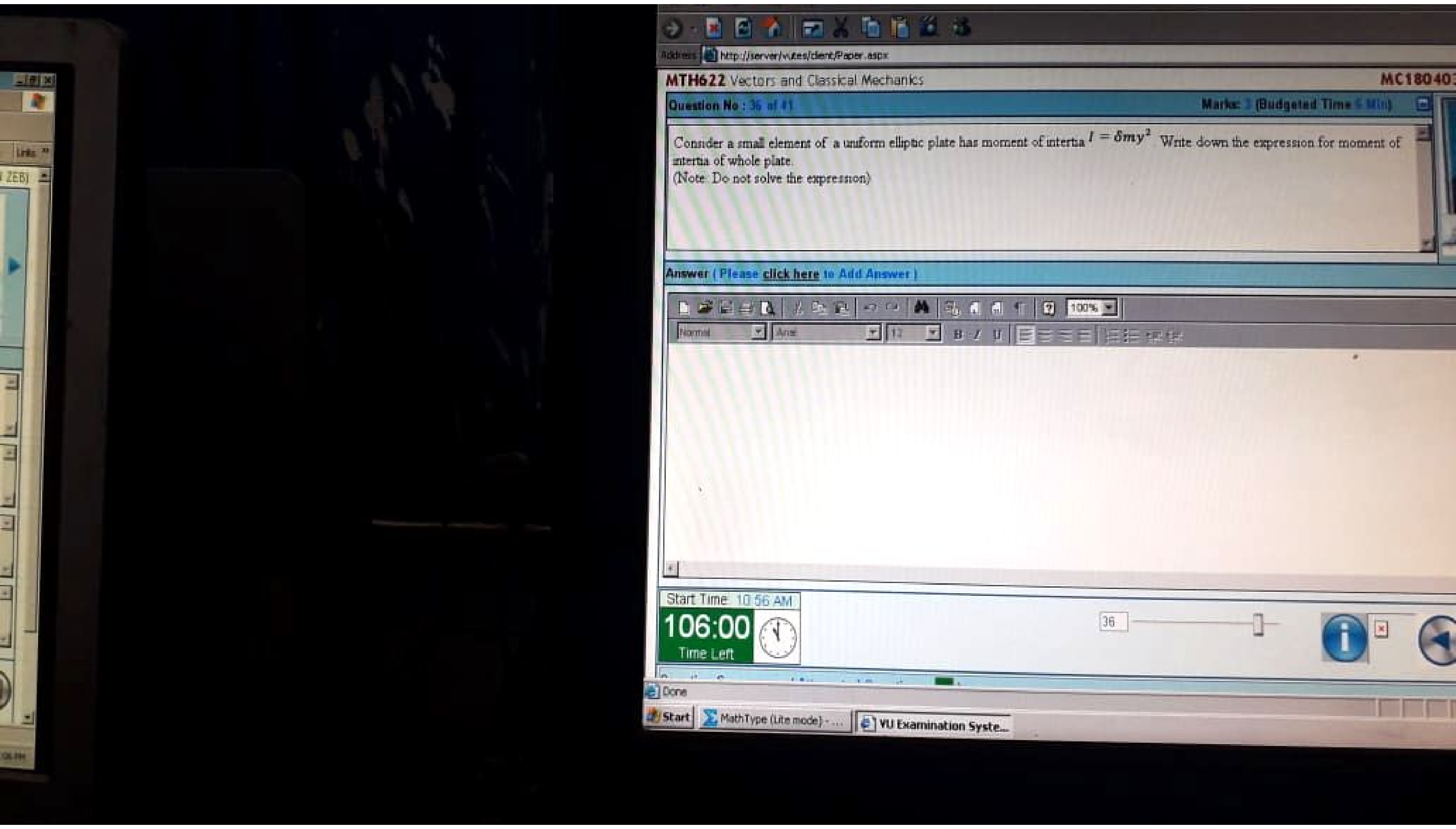
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MTH622 Vectors and Classical Mechanics
Question No : 36 of 41
Marks: 3 (Budgeted Time 6 Min)
Consider a small element of a uniform elliptic plate has moment of intertia $I = \delta m y^2$ Write down the expression for moment of intertia of whole plate.
(Note: Do not solve the expression)
Answer (Please click here to Add Answer)
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MTH622 Vectors and Classical Mechanics

MC1804

Question No : 37 of 41

Marks: 3 (Budgeted Time 6 Min)

If a coordinate system rotates with angular velocity $\vec{\omega} = 2\hat{\imath} - 3\hat{\jmath} + 5\hat{k}$ relative to the fixed coordinate system, and $\vec{A} = \sin t\,\hat{\imath} - \cos t\,\hat{\jmath} + e^{-t}\hat{k}$ then find dA/dt with respect to the fixed point system.

Where $\left(\frac{d\vec{A}}{dt}\right)_r = -\sin t\,\hat{\imath} + \cos t\,\hat{\jmath} - 2e^{-2t}\hat{k}$ and $\vec{\omega} \times \vec{A} = (4e^{-2t} - 3\cos t)\hat{\imath} - (-2e^{-t} + 3\sin t)\hat{\jmath} + (\cos t + 2\sin t)\hat{k}$

Answer (Please click here to Add Answer)

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MTH622 Vectors and Classical Mechanics

Question No : 38 of 41 Marks: 3 (Budgeted Time 6 Min)

What is the impulse developed by a mass of 100 gm which changes its velocity from $i+j+k$ to $2i+2j+3k$ m/sec ?

Answer (Please click here to Add Answer)

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MTH622 Vectors and Classical Mechanics

MC180403

Question No : 32 of 41

Marks: 2 (Budgeted Time 4 Min)

To find the inertia matrix of uniform solid cuboid (parallelepiped) at one of its corner, if $I_{xx} = \frac{M}{3}(b^2 + c^2)$

Then find $I_{yy}, and\ I_{zz}$

Answer (Please **click here** to Add Answer)

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MTH622 Vectors and Classical Mechanics MC18

Question No : 33 of 41 **Marks: 2 (Budgeted Time 4 Min)**

If moment of inertia of a uniform circular cylinder of length b and radius a about x- axis through the center and perpendicular to the central axis is:

$$I_x = m\left(\frac{1}{4}a^2 + \frac{1}{12}b^2\right)$$

Answer (Please click here to Add Answer)

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MTH622 Vectors and Classical Mechanics

Question No : 34 of 41 **Marks: 2 (Budgeted Time 4 Min)**

Write the expression for total energy of a particle executing Simple Harmonic Motion.

Answer (Please **click here** to Add Answer)

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MTH622 Vectors and Classical Mechanics **MC1804**

Question No : 35 of 41 **Marks: 3 (Budgeted Time 6 Min)**

Two objects push off against one another starting from a stationary position. First object with mass 65 kg acquires a velocity of 2 m/s to the right. What velocity does the second object having mass 70 kg acquire?

Answer (Please click here to Add Answer)

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MTH622 Vectors and Classical Mechanics

Question No : 28 of 41

If there is no net force acting on body, then its acceleration is

Answer (Please select your correct option)

- ( ) zero
- ( ) constant
- ( ) increasing
- ( ) decreasing

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MTH622 Vectors and Classical Mechanics

Question No : 29 of 41

Total amount of mass and energy together in a system is

Answer (Please select your correct option)

- increasing
- decreasing
- zero
- constant

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MTH622 Vectors and Classical Mechanics

Question No : 30 of 41

Measure of average kinetic energy of molecules is

Answer (Please select your correct option)

- temperature
- energy
- internal energy
- enthalpy

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MTH622 Vectors and Classical Mechanics

Question No : 31 of 41

What is a degrees of freedom of the system of rigid bodies.

Answer (Please click here to Add Answer)

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MTH622 Vectors and Classical Mechanics

Question No : 27 of 41

Until a force acts on a body, it's velocity is

Answer (Please select your correct option)

- zero
- constant R
- increasing
- decreasing

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MTH622 Vectors and Classical Mechanics

Question No : 26 of 41

The time taken by the particle to complete one oscillation is called

Answer (Please select your correct option)

- amplitude
- time period R
- frequency
- none of these

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MTH622 Vectors and Classical Mechanics

Question No : 25 of 41

The path of projectile is called its

Answer (Please select your correct option)

- velocity
- acceleration
- trajectory R
- none of these

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MTH622 Vectors and Classical Mechanics

Question No : 24 of 41 **Marks: 1 (Budgeted Time**

In a conservative field, the total energy of particle ________________ throughout the motion.

Answer (**Please select your correct option**)

- varies rapidly
- does not remain constant
- remains constant R
- none of these

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MTH622 Vectors and Classical Mechanics

Question No : 20 of 41

Work done is said to be zero, when

Answer **(Please select your correct option)**

- a. Some force acts on a body, but displacement is zero
- b. No force acts on a body but some displacement takes place
- Either a) or b)
- None of these

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MTH622 Vectors and Classical Mechanics

Question No : 21 of 41

The rate of change of displacement of a body is called

Answer (Please select your correct option)

- Velocity
- Acceleration
- Momentum
- None of these

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MTH622 Vectors and Classical Mechanics

Question No : 22 of 41

For an irrotational flow

Answer (Please select your correct option)

- $\nabla \times \vec{v} = \vec{0}$
- $\nabla \times \vec{v} \neq \vec{0}$
- $\nabla . \vec{v} = \vec{0}$
- None of these

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MTH622 Vectors and Classical Mechanics

Question No : 23 of 41

The path of a projectile in a non-resistive medium is

Answer (Please select your correct option)

- linear
- Circular
- Parabolic
- Hyperbolic

R

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Question No : 19 of 41 Marks: 1 (Budgeted Time 1 Min)

The periodic time of a particle with simple harmonic motion is __________ proportional to the angular velocity

Answer (Please select your correct option)

- ( ) Directly R
- ( ) Inversely X
- ( ) Square root
- ( ) None of these

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MTH622 Vectors and Classical Mechanics

Question No : 18 of 41

Marks: 1 (Budgeted Time 1 Min)

For a conservative holonomic dynamical system, the Lagrangian L, kinetic energy T and potential energy V are connected by

Answer (Please select your correct option)

- $L = T + V$
- $L = T - V$
- $L = 2T + V$
- $L = 2T - V$

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MTH622 Vectors and Classical Mechanics

Question No : 17 of 41

At pivot point, body with mass 'm' will have a torque by the

Answer (Please select your correct option)

- force R
- circle
- central point
- radius of circle

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MTH622 Vectors and Classical Mechanics

Question No : 16 of 41 **Marks: 1 (Budgeted Time 1 Min)**

According to parallel axis theorem, the moment of inertia of a section about an axis parallel to the axis through center of gravity (i.e. IP) is given by _______
(where, A = Area of the section, IG = Moment of inertia of the section about an axis passing through its C.G., and h = Distance between C.G. and the parallel axis.)

Answer (Please select your correct option)

- IP = IG + Ah^2
- IP = IG - Ah^2
- IP = IG / Ah^2
- IP = Ah^2 / IG

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MTH622 Vectors and Classical Mechanics

Question No : 13 of 41

Cylinder has moment of inertia given by formula of

Answer (Please select your correct option)

- (ml^2)
- mr
- $\frac{1}{2}$(mr)
- $5(mr^2)$

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MTH622 Vectors and Classical Mechanics

Question No : 14 of 41

Moment of inertia of sphere is given as

Answer (Please select your correct option)

- $2(ml^2)$
- r^2
- $^1/_2(r)$
- $^2/_5(mr^2)$ R

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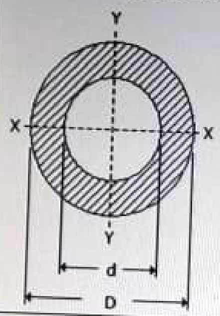
MTH622 Vectors and Classical Mechanics

MC1804

Question No : 15 of 41

Marks: 1 **(Budgeted Time** 1 Min**)**

Moment of inertia of a hollow circular section, as shown in the below figure about X-axis, is



Answer (Please select your correct option)

- $\pi/16\,(D^2 - d^2)$
- $\pi/16\,(D^3 - d^3)$
- $\pi/16\,(D^4 - d^4)$
- $\pi/64\,(D^4 - d^4)$

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MTH622 Vectors and Classical Mechanics

Question No : 12 of 41

Moment of inertia of thin rod is given by formula of

Answer (Please select your correct option)

- $^1/_{12}(ml^2)$
- mr^3
- $2(mr)$
- $^2/_5(mr^2)$

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MTH622 Vectors and Classical Mechanics MC180

Question No : 11 of 41 **Marks: 1 (Budgeted Time 1 Min)**

The moment of inertia of a circular section of base 'b' and height 'h' about an axis passing through its vertex and parallel to base is

Answer (Please select your correct option)

- $\frac{bh^3}{12}$
- $\frac{bh^3}{36}$
- $\frac{bh^3}{3}$
- $\frac{bh^3}{4}$ R

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MTH622 Vectors and Classical Mechanics

Question No : 10 of 41

The wheels of a moving car possess

Answer (Please select your correct option)

- Potential energy only
- Kinetic energy of translation only
- Kinetic energy of rotation only
- Kinetic energy of translation and rotation both

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MTH622 Vectors and Classical Mechanics

Question No : 9 of 41

The polar moment of inertia of a circular section is about

Answer (Please select your correct option)

- X-X axis
- Y-Y axis
- Z-Z axis ✓ R
- Neutral axis

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MTH622 Vectors and Classical Mechanics

Question No : 5 of 41

The functions in Green's theorem will be

Answer (Please select your correct option)

- Continuous derivatives
- Discrete derivatives
- Continuous partial derivatives R
- Discrete partial derivatives

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MTH622 Vectors and Classical Mechanics MC18

Question No : 6 of 41 **Marks:** 1 (Budgeted Time 1 Min)

If a time dependent vector function $\vec{A}$ is represented by $\vec{A}_f$ and $\vec{A}_r$ in fixed and rotating coordinate system, then

Answer (Please select your correct option)

- $\left(\frac{d\vec{A}}{dt}\right)_f = \left(\frac{d\vec{A}}{dt}\right)_r - \omega \times \vec{A}_r$
- $\left(\frac{d\vec{A}}{dt}\right)_f = \left(\frac{d\vec{A}}{dt}\right)_r + \omega \times \vec{A}_r$ R
- $\left(\frac{d\vec{A}}{dt}\right)_f = A_r - \omega \times \vec{A}_r$
- $\left(\frac{d\vec{A}}{dt}\right)_f = \left(\frac{d\vec{A}}{dt}\right)_r - A \times \vec{A}_r$

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MTH622 Vectors and Classical Mechanics

Question No : 7 of 41

The axes relative to which product of inertia are zero are called

Answer (Please select your correct option)

- principal axes
- principal moment of inertia
- moment of inertia
- axes of rotation

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MTH622 Vectors and Classical Mechanics

Question No : 8 of 41 **Marks:** 1 (Budgeted Time 1 Min)

A 3×3 symmetrical matrix has ------------- real eigenvalues, which may be distinct or repeated

Answer (Please select your correct option)

- Infinite
- Zero
- Two
- Three

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MTH622 Vectors and Classical Mechanics

Question No : 4 of 41

The Stoke's theorem can be used to find which of the following?

Answer (Please select your correct option)

- Area enclosed by a function in the given region
- Volume enclosed by a function in the given region
- Linear distance
- Curl of the function

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MTH622 Vectors and Classical Mechanics

Question No : 3 of 41

$\nabla \times \nabla \varphi = ?$

Answer (Please select your correct option)

- $\nabla^2 \varphi$
- φ
- 0
- None of these

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MTH622 Vectors and Classical Mechanics

Question No : 2 of 41 **Marks: 1 (Budgeted Time 1**

Which of the following property is not true?

Answer (Please select your correct option)

- $\nabla.(\vec{A}+\vec{B}) = \nabla.\vec{A} + \nabla.\vec{B}$
- $\nabla.(\varphi\vec{A}) = \varphi(\nabla.\vec{A}) + \vec{A}.(\nabla\varphi)$
- $\nabla.\nabla\varphi = \nabla^2\varphi$
- $\nabla.\vec{V} = \vec{V}.\nabla$

R

MTH622 Vectors and Classical Mechanics

Question No : 1 of 41 MC18

Marks: 1 (Budgeted Time 1 Min)

The maximum value of the directional derivative of $\varphi(x, y, z)$ is equal to the

Answer (Please select your correct option)

- Magnitude of $\nabla \times \varphi$
- Magnitude of φ
- Magnitude of $\nabla \varphi$ ✓ R
- None of these